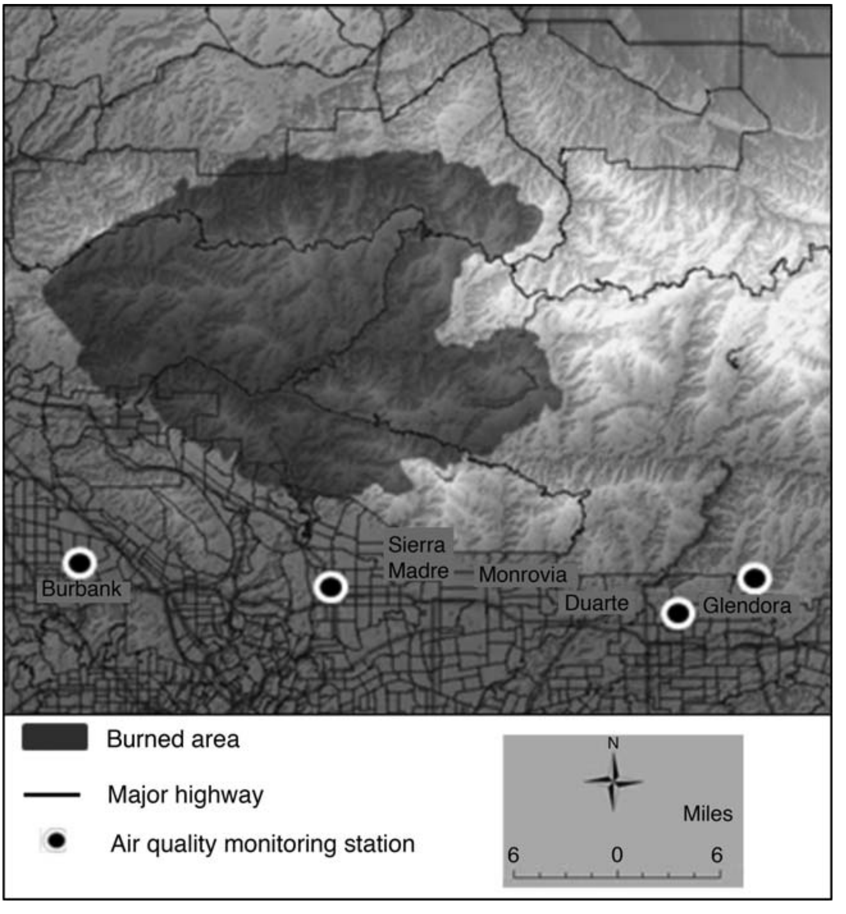

FIGURE 1
Location of Station Fire, Cities Surveyed, and Monitoring Stations
Adapted from: http://seamless.usgs.gov/index.php.

Four respondents reported defensive expenditures well above the sample mean, so any expenditure from these four respondents greater than 3 standard deviations from the sample mean was recoded to the highest value without the outlier.

To facilitate the quantification of a COI estimate for each individual, only those respondents who experienced health symptoms as a result of exposure to the wildfire smoke were asked an additional set of questions about the medical care they received and the private, out-of-pocket costs incurred for this treatment. They were also asked to report the time spent traveling to, waiting for, and receiving any medical care. In addition, respondents were asked to report any expenditure on nonprescription medications taken to alleviate symptoms, as well as the number of work and/or recreation days missed as a direct result of these symptoms. Elicited in this way, these costs of illness represent an exogenous outcome of the illness experienced and are thus distinct from defensive actions, which are chosen by the individual. This distinction has been the source of some confusion in the literature, as explained in Bockstael and McConnell (2007). Table 3 outlines the number and percentage of survey respondents who undertook these remedial actions, along with

TABLE 1
Objective and Subjective Pollution Levels by City during the Station Fire

City	CO (ppm)		PM2.5 ($\mu g/m^3$)		PM10 ($\mu g/m^3$)		Smelled Smoke Inside of Home			Smelled Smoke Outside of Home		
	Average (6-d mean)	Peak (6-d mean)	Average (6-d mean)	Peak (6-d mean)	Average (6-d mean)	Peak (6-d mean)	None	1–5 days	> 5 days	None	1–5 days	> 5 days
Duarte ($n = 54$)	0.68	1.40					18 (33%)	20 (37%)	16 (30%)	2 (4%)	15 (28%)	37 (69%)
Monrovia ($n = 84$)	0.68	1.40					29 (35%)	29 (35%)	26 (31%)	2 (2%)	24 (29%)	58 (69%)
Sierra Madre ($n = 31$)	0.48	1.80					16 (52%)	9 (29%)	6 (19%)	3 (10%)	9 (29%)	19 (61%)
Burbank ($n = 80$)	0.64	1.57	25.18	93.50			37 (46%)	24 (30%)	19 (24%)	4 (5%)	28 (35%)	48 (60%)
Glendora ($n = 164$)	0.65	1.42	46.83	120.83	53.82	133.12	75 (46%)	56 (34%)	33 (20%)	11 (7%)	61 (37%)	92 (56%)

Note: 6-d mean represents six-day averages from the period August 26–August 31, 2009.

the average cost reported. Respondents were also asked a series of questions about their health history, various lifestyle factors, and demographic information.

For the CVM model, only those respondents who experienced health symptoms themselves or had household members experiencing symptoms from the Station Fire smoke were asked an additional question about their WTP to reduce the symptom days their household experienced by 50%. Respondents were told that they would be asked a question about what it would have been worth to them to reduce some of the effects their household may have experienced as a result of wildfire smoke exposure. They were specifically instructed to focus on the health effects experienced, the preventative actions taken to avoid these health effects, as well as days of work and recreation lost as a direct result of smoke from the wildfire when answering the question. While there are many other negative impacts of wildfire, such as damage to the home and amenity effects such as reduced visibility from wildfire smoke, respondents were specifically asked not to consider any of these additional effects. This was done in an effort to make the CVM WTP value as comparable to the DBM WTP value as possible. A dichotomous choice question format was used with 10 different bid amounts ranging from \$10 to \$750 based on focus groups and acute morbidity values from various studies summarized in Dickie and Messman (2004). This type of WTP question format was utilized due to the desirable incentive compatibility properties it has. Survey respondents are presented with only one bid amount and therefore cannot easily influence potential outcomes by purposely revealing values other than their true WTP (Haab and McConnell 2002; Boyle 2003). A summary of all study variables can be found in Table 4. Table 5 identifies the percentage of "yes" responses to the WTP question at each bid amount.

VI. ECONOMETRIC ESTIMATION

Defensive Behavior Model

To implement the DBM to calculate the mean WTP for a reduction in symptom days,

TABLE 2
Defensive Actions Taken by Respondents and Average Expenditure on Each ($N = 413$)

Defensive Actions	Percentage of Survey Respondents	Average Expenditure
Evacuated	5.6%	$257.95
Wore a face mask	7.0%	$6.04
Used a home air cleaner	21.3%	$26.93
Avoided going to work	4.6%	$219.41[a]
Removed ashes from property	57.4%	$8.67
Ran air conditioner more than usual	60.3%	$27.66[b]
Stayed indoors more than usual	73.1%	N/A
Avoided normal outdoor recreation/exercise	77.7%	N/A

Note: N/A, not available.
[a] Lost earnings reported by respondent.
[b] Respondents were not asked to report this cost. The price was calculated as (the kilowatt hours per day used in running the air conditioner) × (the cost per kilowatt hour) × (the average number of days respondents took this defensive action). According to the California Energy Commission, the average California resident uses 27 kW-h to run his or her central air conditioning for 12 hours/day (assuming the air conditioner is run for 120 days of the year). According to the U.S. Energy Information Administration, residents in California in September of 2009 were charged 15.76 cents/kW-h used. Respondents who ran the air conditioning more as a result of the wildfire smoke ran it for an average of 6.5 days. This results in a value of $27.66.

TABLE 3
Remedial Actions Taken by Respondents and Average Expenditure on Each ($N = 413$)

Remedial Actions	Percentage of Survey Respondents	Average Expenditure
Obtained medical care/prescription medications	6.3%	$77.87[a]
Took nonprescription medicines	12.6%	$16.86
Went to a nontraditional healthcare provider	1.2%	$33.00
Missed work	3.6%	$691.76
Missed days of recreation activities	27.6%	N/A

Note: N/A, not available.
[a] Includes the opportunity cost of time spent traveling to and receiving medical care, calculated as (the number of hours spent in these activities) × (the hourly wage rate reported by that respondent).

a health production function such as that in equation [2] is estimated. The health outcome experienced is the dependent variable of interest, which in this case is the number of days that health symptoms were experienced as a direct result of exposure to wildfire smoke. The independent variables include everything that enters the right-hand side of the health production function, including pollution levels, defensive actions taken, the individual's health history, lifestyle factors, and demographics.

Estimating this model has proven somewhat difficult in practice. A major complication that arises in empirical estimation, explained thoroughly by Dickie (2003), is the fact that defensive action variables are often endogenous, jointly determined with the health outcome. These endogenous regressors will be correlated with the disturbance of the health production function equation they appear in, meaning least squares estimators will be both biased and inconsistent. Numerous studies that have estimated health production function regression models have expressed the importance of this issue (Gerking and Stanley 1986; Joyce, Grossman, and Goldman 1989; Alberini et al. 1996; Dasgupta 2004; Dickie 2005).

The dependent variable in this analysis is count in nature (the number of days symptoms were experienced). The potentially endogenous defensive action variables are binary, meaning nonlinear estimation techniques to

TABLE 4
Study Variables and Summary Statistics

Variable	Coding	Mean	Std. Dev.
Perceived Pollution Levels			
Smelled smoke indoors > 5 days	1 = yes, 0 = no	0.24	0.43
Smelled smoke outdoors > 5 days	1 = yes, 0 = no	0.62	0.49
Objective Pollution Levels			
Average daily maximum CO concentration	Parts per million (ppm)	1.47	0.11
Illness Information			
Symptom days	Count	3.28	6.06
Ear, nose or throat symptoms	1 = yes, 0 = no	0.36	0.48
Breathing symptoms	1 = yes, 0 = no	0.18	0.39
Heart symptoms	1 = yes, 0 = no	0.04	0.20
Other symptoms	1 = yes, 0 = no	0.09	0.28
Household symptom days	Count	5.94	9.35
Number of household members with symptoms	Count	0.89	1.24
Averting Actions			
Evacuated	1 = yes, 0 = no	0.06	0.23
Wore a face mask	1 = yes, 0 = no	0.07	0.26
Used a home air cleaner	1 = yes, 0 = no	0.21	0.41
Avoided going to work	1 = yes, 0 = no	0.05	0.21
Removed ashes from property	1 = yes, 0 = no	0.57	0.50
Ran air conditioner more than usual	1 = yes, 0 = no	0.60	0.49
Stayed indoors more than usual	1 = yes, 0 = no	0.73	0.44
Avoided normal outdoor recreation/exercise	1 = yes, 0 = no	0.78	0.42
Mitigating Actions			
Obtained medical care/prescription medications	1 = yes, 0 = no	0.06	0.24
Took nonprescription medications	1 = yes, 0 = no	0.13	0.33
Went to a nontraditional healthcare provider	1 = yes, 0 = no	0.01	0.11
Missed work	1 = yes, 0 = no	0.04	0.19
Missed days of recreation activities	1 = yes, 0 = no	0.28	0.45
Health History			
Current respiratory condition	1 = yes, 0 = no	0.12	0.32
Current heart condition	1 = yes, 0 = no	0.09	0.28
Experienced health effects from wildfire smoke in past	1 = yes, 0 = no	0.24	0.42
Health and Lifestyle			
Times per week of exercise	0 = 0 times/week; 1 = 1–2 times/week; 2 = 3–5 times/week; 3 = > 5 times/week	1.62	0.92
Smoker	1 = yes, 0 = no	0.08	0.28
Alcoholic drinks per week	0 = none; 1 = 1–7 drinks/week; 2 = 8–14 drinks/week; 3 = > 14 drinks/week	0.60	0.73
Current health is excellent	1 = yes, 0 = no	0.29	0.45
Current health is good	1 = yes, 0 = no	0.55	0.50
Current health is fair	1 = yes, 0 = no	0.14	0.35
Current health is poor	1 = yes, 0 = no	0.02	0.14

(table continued on following page)